

Electroplated Cu-Diamond Composite Thermal Vias for Interposer-Level Heat Management in 2.5D/3D Packages

Ye Yang

Department of Mechanical and
Aerospace Engineering,
University of California,
Los Angeles, CA 90095
yangye0646@ucla.edu

James Chien

Taiwan Foresight Co., Ltd.,
Section 1, 1 No.1 Road, Jiafeng,
JhongSing Village, Jhubei City,
Hsinchu County, 302, Taiwan
james@foresight-web.com.tw

Tiwei Wei*

Department of Mechanical and
Aerospace Engineering,
University of California,
Los Angeles, CA 90095
tiwei32@ucla.edu

Abstract—Glass has emerged as a promising interposer material for 2.5D advanced packaging. Its low thermal conductivity, however, remains a critical bottleneck for heat dissipation. Thermal through-glass vias (TGVs) are therefore essential for practical deployment. Current solutions rely on pure copper fill, which offers limited thermal benefit. This work proposes copper-diamond composite as a TGV fill material for glass interposers. A fabrication process combining vacuum-assisted diamond particle injection with two-step double-sided electroplating was developed. Fully filled Cu-diamond composite TGVs were demonstrated in a 1000 μm thick glass core with 100 μm diameter vias, achieving an aspect ratio of 10:1. Cross-sectional analysis confirmed void-free filling and uniform diamond particle distribution. This work presents a first demonstration of high-aspect-ratio Cu-diamond composite TGVs. It offers a new route toward improved thermal management in glass-based 2.5D packaging.

Keywords—Through-glass via (TGV), Glass interposer, Copper-diamond composite, Thermal management, 2.5D packaging, Electroplating

I. INTRODUCTION

The semiconductor industry is moving beyond monolithic System-on-Chip (SoC) designs. This shift has accelerated heterogeneous integration. In heterogeneous integration, chiplets made at different process nodes are co-packaged to optimize power, performance, area, and cost [1], [2]. Among these packaging options, 2.5D integration with passive interposers is especially attractive for bandwidth-intensive systems such as artificial intelligence (AI) accelerators and high-performance computing. These systems require dense die-to-die interconnects at fine pitch [1]. In a typical 2.5D layout, multiple chiplets are placed side by side on a shared interposer. This structure provides interconnect densities far beyond conventional printed circuit boards, and also allows logic and memory to be manufactured separately and then integrated at the package level.

Silicon interposers with through-silicon vias (TSVs) are the established solution for 2.5D integration. However, they are limited by high fabrication cost and parasitic substrate loss [3]. Glass has emerged as an alternative interposer material. Glass offers high electrical resistivity, a low dielectric constant, and a

tunable CTE that can be closely matched to silicon. These properties reduce signal insertion loss at high frequencies and lower redistribution layer (RDL) capacitance compared with silicon substrates [3], [4], [5]. Glass is also compatible with large-panel processing, which can further reduce manufacturing cost [6]. As a result, glass interposers have been widely studied. Prior work has demonstrated Cu-filled TGV arrays, fine-pitch RDL formation, and full package assembly [7], [8].

A key challenge is thermal management. The thermal conductivity of glass ($\kappa \approx 1 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) is nearly two orders of magnitude lower than that of silicon ($\kappa \approx 150 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) [9]. As chip power density increases, heat removal through the glass core becomes a major bottleneck. Simulations show that increasing interconnections arrays can reduce junction temperatures in glass interposer. This can bring thermal performance close to that of silicon interposer systems [9]. For this reason, thermal through-glass vias (TGVs) are considered a necessary element in practical glass interposer designs [10].

Today, TGV metallization relies on copper electroplating. Bottom-up Cu superconformal filling in high-aspect-ratio vias has been demonstrated using specialized additive systems. Examples include thiazolyl blue tetrazolium bromide (MTT) and TNBT/NTBC combinations. These additives suppress sidewall deposition while accelerating bottom growth [11], [12]. Multi-step plating and pulse-periodic-reverse (PPR) electroplating can further improve Cu uniformity, grain structure, and ductility in TGVs [13], [14], [15]. After plating, heat treatment is commonly used to reduce residual stress, suppress Cu protrusion caused by CTE mismatch between copper and glass, and stabilize the microstructure [16]. Despite these advances, existing TGV filling technology is limited to pure copper ($\kappa \approx 400 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$). No higher-conductivity composite fill material has been reported for glass interposer vias.

Copper-diamond composites are widely studied for thermal management. They combine copper's processability with diamond's very high thermal conductivity ($\kappa = 1,200\text{--}2,000 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) and low CTE ($\approx 1 \times 10^{-6} \text{ K}^{-1}$) [17], [18]. Many fabrication routes have been reported. Early work used high-temperature high-pressure (HTHP) sintering and showed that composite thermal conductivity depends on diamond particle

size and volume fraction [19]. Spark plasma sintering (SPS) uses pulsed joule heating to densify material at lower temperatures. Pan et al. reported $609 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ for ZrC/Zr bilayer-coated diamond at 60 vol% loading [18]. Melt infiltration and liquid-solid separation can infiltrate a diamond preform with liquid copper and form near-net-shape parts. Using large particles, thermal conductivity up to $552 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ has been reported [20]. In all of these approaches, interfacial engineering is critical. Liquid copper wets bare diamond poorly, with a contact angle above 130° at elevated temperatures. This leads to large thermal boundary resistance ($\text{TBR} > 10^{-7} \text{ m}^2\cdot\text{K}\cdot\text{W}^{-1}$) and reduces effective thermal conductivity [18]. Carbide-forming interlayers such as TiC, and WC can reduce TBR by promoting stronger bonding at the Cu-diamond interface. These layers are commonly deposited by magnetron sputtering or electroless plating [18], [21].

Electrodeposition provides a room-temperature and low-cost alternative. Susumu et al. formed Cu/diamond composites by potentiostatic electroplating after sedimenting diamond particles on the cathode. They reported $600 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ at about 49 vol% diamond [22]. Adding TiC coatings on diamond particles increased thermal conductivity to $557 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ at only 34.7 vol%. This value is about 40% higher than pure copper, and the improvement was attributed to better interfacial thermal conductance [23]. Wu et al. reported electroplated composites with near-perfect Cu-diamond interfaces and a thermal conductivity of $615 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ under optimized current density [24]. A later three-step electroplating process (substrate plating, diamond embedding, and copper overcoating) produced 262 μm thick single-layer composite sheets with $588 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ at 36 vol% diamond [25]. In parallel, time-domain thermoreflectance (TDTR) has been used to measure the thermal conductivity of individual diamond microparticles. Recent reports show strong size dependence, from $\sim 400 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ for $20 \mu\text{m}$ particles to over $2000 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ for $300 \mu\text{m}$ particles [26], [27]. Thermoreflectance metrology has further advanced the characterization of metal-diamond heat spreaders [28].

Despite this progress in bulk Cu-diamond composites and Cu electroplating for TGVs, no prior study has demonstrated Cu-diamond composite fill in through-via structures of any kind. Existing bulk fabrication methods, including sintering, melt infiltration, and planar electroplating, do not fit the confined, high-aspect-ratio geometry of through-holes in glass interposers. This paper addresses this gap. We develop a vacuum-assisted diamond particle injection process and combine it with a two-step double-sided electroplating strategy [29]. Using this approach, we achieve fully filled Cu-diamond composite TGVs with a 10:1 aspect ratio in a $1000 \mu\text{m}$ thick glass core. Cross-sectional analysis shows void-free filling and a uniform diamond particle distribution. This is the first demonstration of Cu-diamond composite fill in glass interposer through-vias. The result provides a new route for thermal management in glass-based 2.5D packaging.

II. PARTICLE SIZE SELECTION: THEORETICAL BASIS

A. Geometric Constraint

To prevent particle bridging across the via entrance, the particle diameter was selected such that $d_p \leq D_{\text{via}}/5$, consistent with empirical bridging criteria established in particulate flow

and powder injection molding. For a $100 \mu\text{m}$ via, this imposes a hard geometric upper limit:

$$d_p \leq \frac{D_{\text{via}}}{5} = 20 \mu\text{m}$$

B. Thermally Effective Range (Hasselman–Johnson Model)

The effective thermal conductivity of a Cu-diamond composite is estimated using the Hasselman–Johnson (HJ) model [17]:

$$k_{\text{eff}} = k_m \frac{k_p(1 + 2\alpha) + 2k_m + 2V_p[k_p(1 - \alpha) - k_m]}{k_p(1 + 2\alpha) + 2k_m - V_p[k_p(1 - \alpha) - k_m]}$$

where $\alpha = k_m/(h_c r_p)$ is the dimensionless interface resistance parameter. Parameter used: $k_m = 400 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$, $h_c = 1.25 \times 10^8 \text{ W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$ (interface thermal conductance for TiC-coated Cu-diamond was determined from Cho et al. [23]), $V_p = 0.50$. Setting $k_{\text{eff}} = k_m$ yields the critical particle diameter below which diamond addition is thermally counterproductive:

$$d_{\text{crit}} = \frac{2k_p k_m}{h_c(k_p - k_m)}$$

Using measured size-dependent diamond particle conductivity values ($k_p = 774.6 - 1366 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ [27]), d_{crit} falls between 9.1 and $13.2 \mu\text{m}$, as shown in Fig. 1.

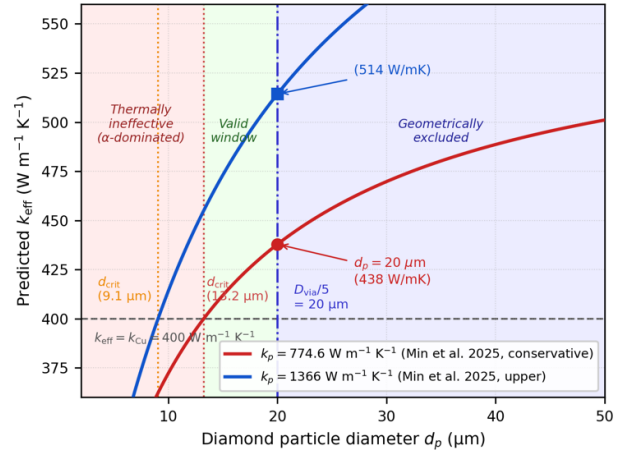


Fig. 1 Effective thermal conductivity vs. diamond particle diameter from the Hasselman–Johnson model ($V_p = 0.50$). Shaded regions mark the thermally ineffective ($d_p < d_{\text{crit}}$) and geometrically excluded ($d_p > 20 \mu\text{m}$) regimes.

C. Design point

The $20 \mu\text{m}$ particle diameter simultaneously satisfies both constraints: it is the geometric maximum for void-free via filling and lies above the thermally effective threshold. At this design point, the HJ model predicts $k_{\text{eff}} = 438 - 514 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ (9.5–28.6% above pure copper), depending on the assumed k_p . This provides the theoretical basis for the process developed in Section III.

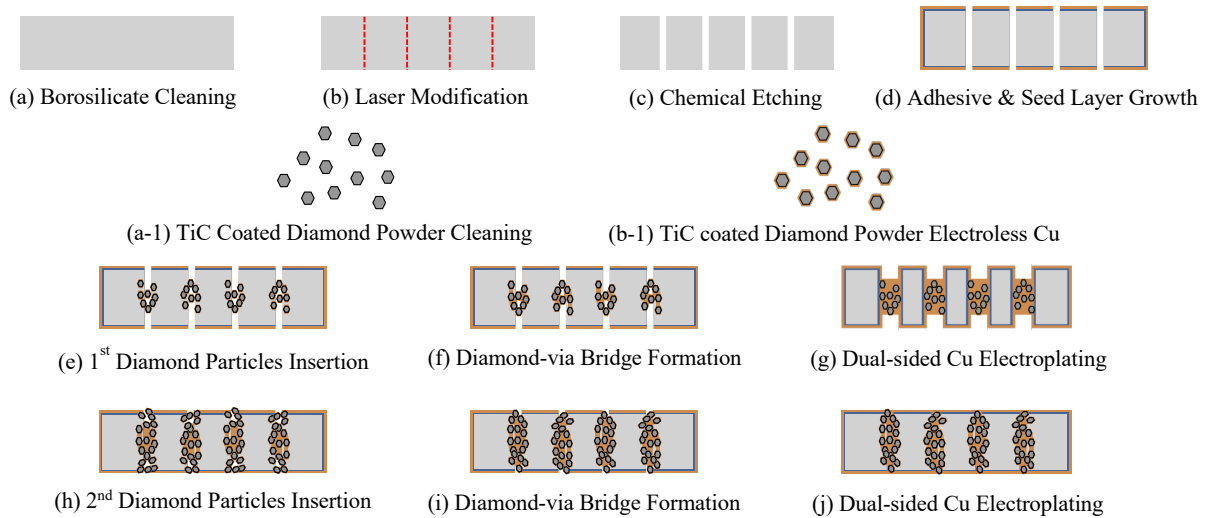


Fig. 2 Fabrication process flow of Cu-diamond composite thermal vias.

III. NOVEL CU-DIAMOND COMPOSITE THERMAL VIA FABRICATION PROCESS

The fabrication flow is shown in Fig. 2. It combines glass core preparation and diamond particle treatment. The full process has ten steps. The goal is to obtain void-free, high-density Cu–diamond composite filling in high-aspect-ratio TGVs.

The glass process starts with cleaning the borosilicate wafer. The wafer then undergoes laser modification. This step creates subsurface damage. The damaged region etches faster, so chemical etching can form vertical through glass vias. After the vias are formed, a Ti adhesion layer and a Cu seed layer are deposited by sputtering. These layers make the via sidewalls electrochemically active for later plating. In parallel, TiC pre-coated diamond particles are prepared. The particles are cleaned and then encapsulated with copper by electroless deposition. This copper shell improves thermal interface compatibility. It also allows the particles to bond electrochemically to the via interior.

Via filling is performed in two identical cycles. In the first cycle, the Cu-encapsulated diamond particles are injected into the vias under vacuum assistance. The pressure difference pushes particles into the 1000 μm -deep channels. Immediately after injection, a short electroless Cu step is applied. During this step, the copper shell on each particle bonds to the Cu seed layer on the via sidewalls. This bonding anchors the particles and reduces particle movement during the later electroplating step. After anchoring, dual-sided Cu electroplating is used to consolidate the first particle layer. The second cycle repeats the same sequence: injection, anchoring, and electroplating. This second cycle increases the particle density and improves uniformity along the via depth. It also reduces settling and agglomeration that can occur if all particles are injected in a single high-concentration step. This two-cycle strategy is essential to reach the target loading of 50 vol% while keeping the particle distribution homogeneous throughout the via.

A. Diamond Particle Surface Modification

The diamond particles were purchased from Element Six Inc. The deposited TiC coating layer of the diamond has two roles. First, it lowers the thermal resistance at the Cu–diamond interface by introducing a carbide interlayer with covalent bonding, instead of a van der Waals contact. Second, it provides a surface that is suitable for electroless metal deposition.

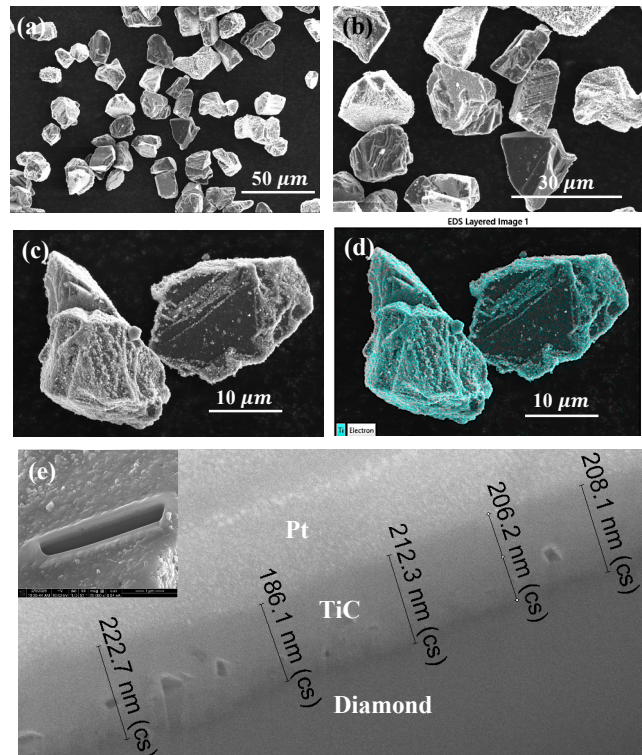


Fig. 3 TiC-coated diamond particles. (a–c) SEM images at different magnifications. (d) Ti EDS overlay indicates surface coverage. (e) FIB cross-section shows a ~200 nm TiC coating on diamond (Pt protection cap on top).

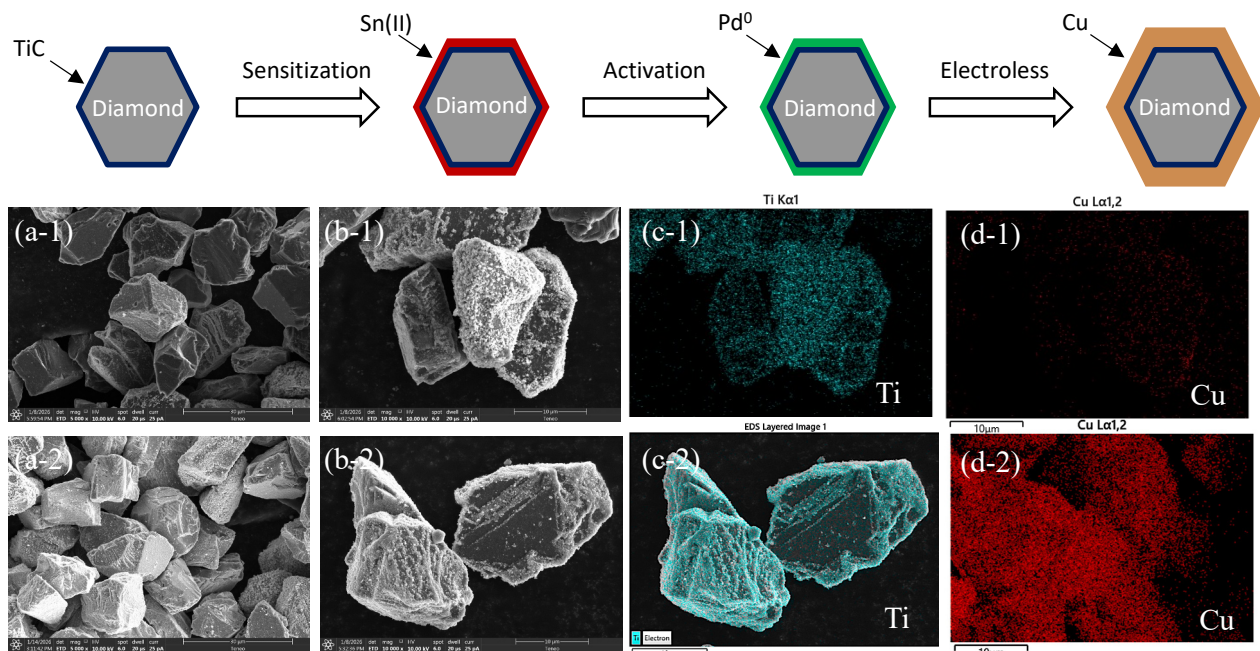


Fig.4 TiC-coated diamond particles copper electroless plating (SLE): (a-1)-(d-1) electroless copper deposition 2 minutes; (a-2)-(d-2) electroless copper deposition 30 minutes.

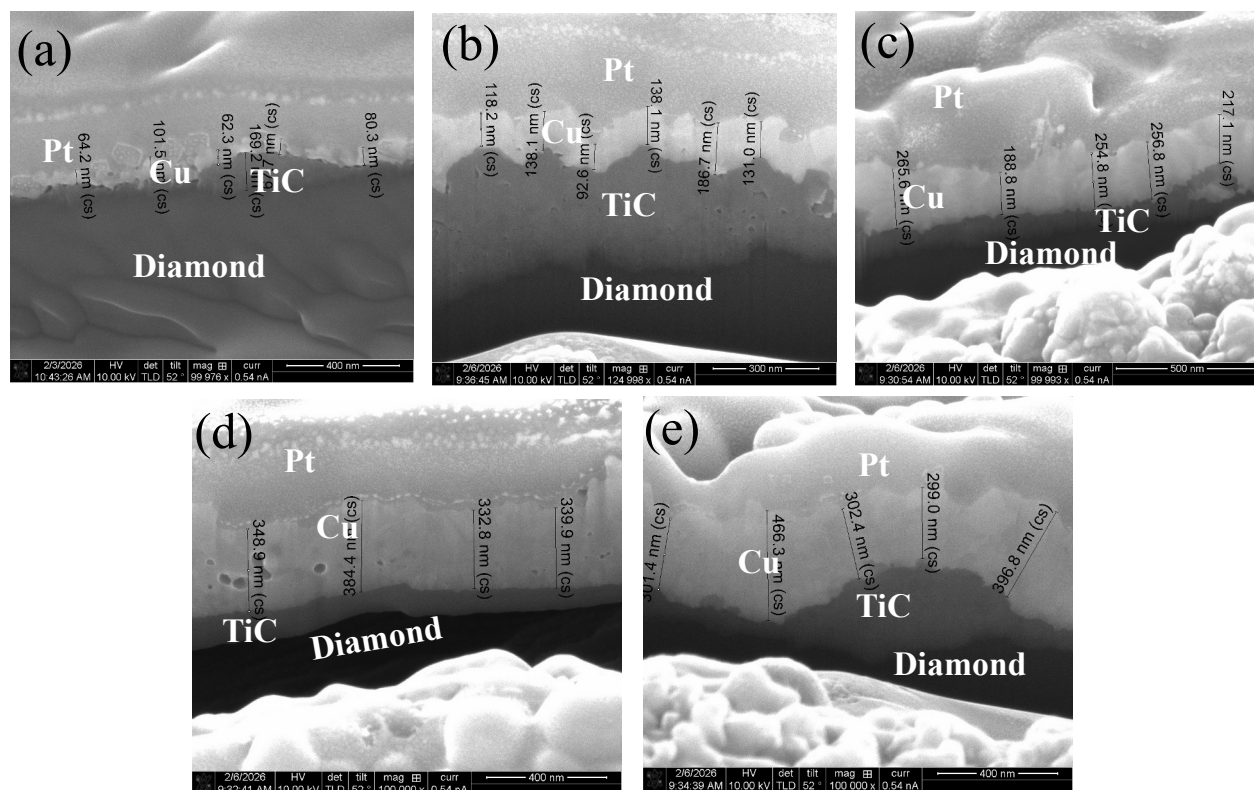


Fig.5 SEM images of diamond particles after electroless copper deposition at different plating times: (a) 2 min, (b) 10 min, (c) 15 min, (d) 30 min, and (e) 60 min.

Cross-sectional SEM images of the as-received particles from Fig.3 show a conformal TiC layer. The coating thickness is ~ 200 nm and is uniform over the particle surface.

Cu shell was deposited on the TiC-coated diamond to ensure electrochemical compatibility with the Cu seed layer inside the via. A three-step electroless process was used: (i) sensitization in SnCl_2 , (ii) activation in PdCl_2 , and (iii) time-controlled electroless Cu deposition for seed layer enhancement. The sensitization and activation steps form catalytic Pd nuclei on the TiC surface. These nuclei initiate autocatalytic Cu deposition during the enhancement step. From Fig. 4, we can see that different seed layer enhancement times lead to different thicknesses of the outermost copper layer. When the electroless plating time is 2 min, there is almost no visible copper coverage on the diamond particle surface, and the signal is very weak. In contrast, when the electroless plating time is extended to 30 min, a much stronger copper signal is observed under the same measurement conditions, indicating that the diamond particles are fully covered by a continuous copper layer.

To systematically study the relationship between electroless plating time and the thickness of the outer copper layer on the diamond particles, we tested five reaction durations: 2, 10, 15, 30, and 60 min.

Based on Fig. 5, we can qualitatively observe that the Cu thickness on the outer surface of the diamond particles increases with plating time. Therefore, we performed measurements to determine the specific thickness values of the outer copper layer.

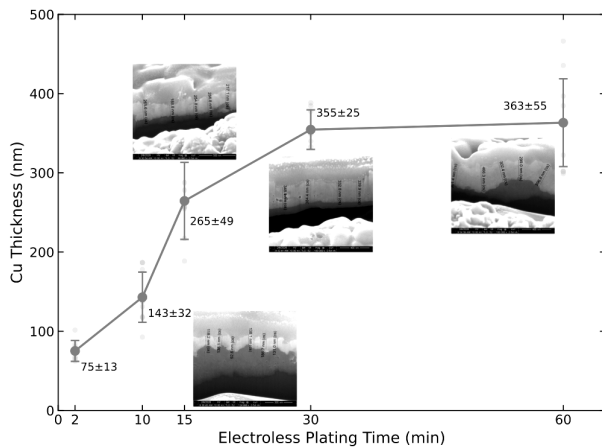


Fig. 6 Copper coating thickness on TiC-coated diamond particles as a function of electroless plating time. Data points represent individual measurements, and error bars indicate standard deviation.

Fig. 6 shows the Cu thickness increases from 75 nm at 2 min to 355 nm at 30 min. Growth slows after 30 min and approaches saturation (363 nm at 60 min). This behavior is attributed to the increasing diffusion barrier created by the growing Cu layer, which limits precursor transport to the catalytic surface. Based on these results, a 30 min enhancement time was used in all subsequent via-filling experiments. This condition provides a continuous Cu shell while avoiding unnecessary processing time.

B. Cu-diamond Composite Thermal Via Filling

Before via filling, the electroplating conditions were verified using planar Cu–diamond composite samples. Cu-encapsulated diamond particles were placed on a Cu-seeded substrate and consolidated by Cu electroplating. Identical electrolyte composition has been adopted.

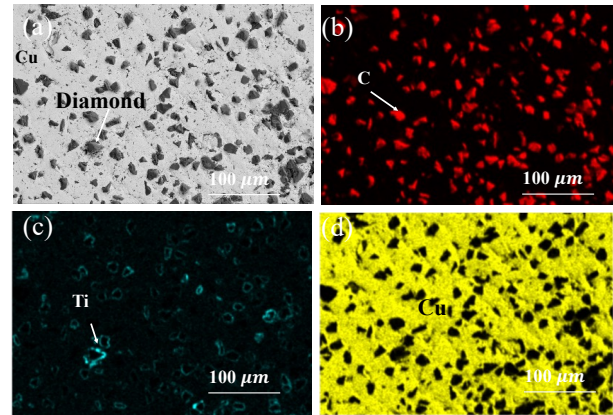


Fig. 7 Cross-sectional microstructural and compositional characterization of the planar plated Cu–diamond composite: (a) SEM image, and corresponding EDS elemental maps of (b) C, (c) Ti, and (d) Cu.

Fig. 7 shows a cross-sectional SEM image of the plated planar sample. The composite layer is densely consolidated and void-free. The diamond particles are fully embedded in the Cu matrix, and no interfacial delamination is observed. This result confirms that a 30-min electroless Cu encapsulation produces sufficient electrical continuity for uniform galvanic deposition. However, unlike planar electroplating where a constant current density can be maintained throughout the process, TGV filling requires a progressive current ramping strategy to accommodate the evolving via fill geometry. The planar experiments served to validate the electroless encapsulation quality and baseline electrolyte compatibility, while the current profile for TGV filling was independently optimized to ensure stable bottom-up growth from both via entrances, as described below.

In terms of dual-sided via filling, after the two-cycle particle injection and electroless anchoring procedure described previously, electroplating was performed on 1000 μm -thick borosilicate glass substrates with 100 μm -diameter TGVs (aspect ratio 10:1) from both sides. Bottom-filling growth was initiated from both via entrances to mitigate voids and defects that commonly occur in single-sided filling at this aspect ratio. The optimization process involves vacuum conditions and injection time, resulting in different diamond particle filling ratios, as shown in Fig. 8.

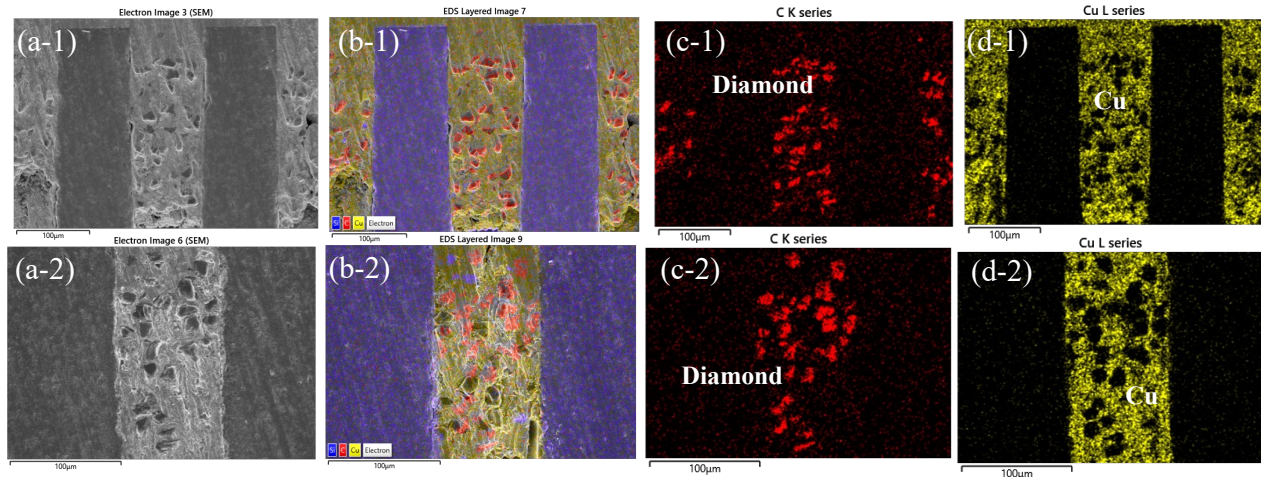


Fig.8 Cross-sectional microstructural and compositional characterization of the Cu–diamond composite within the TGV under different injection times during the optimization process: (a) single cycle injection; (b) double cycle injection.

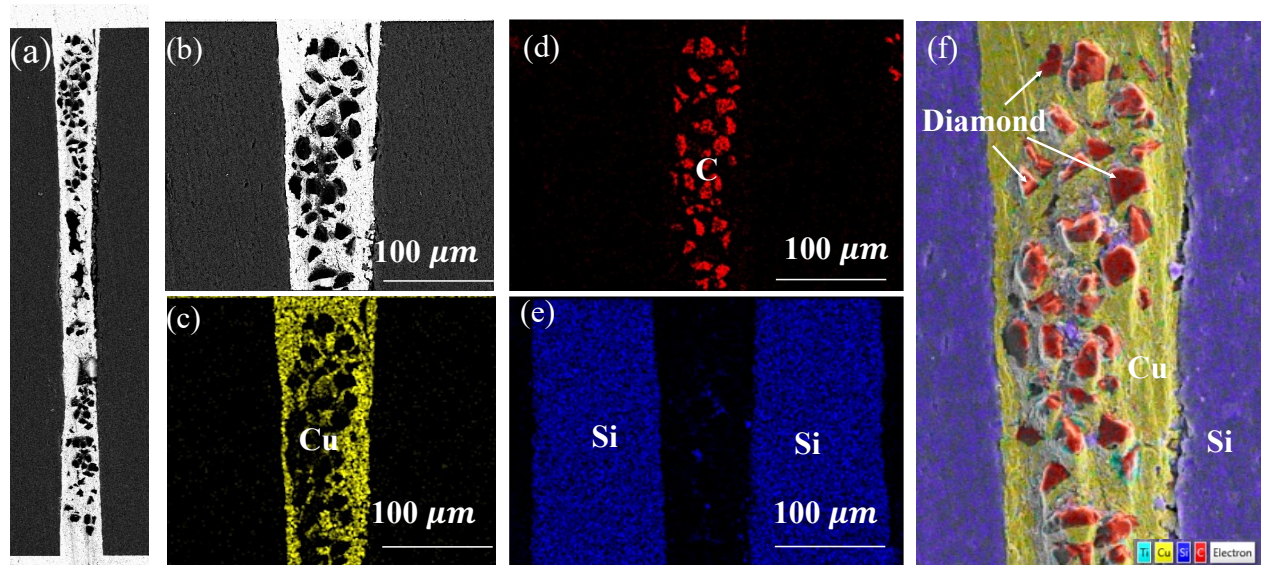


Fig.9 Cross-sectional microstructural and compositional characterization of the Cu–diamond composite within the TGV: (a) low-magnification SEM image of the via, (b) higher-magnification SEM image of the composite filling, and corresponding EDS elemental maps of (c) Cu, (d) C, (e) Si, and (f) EDS map for all elements in TGV.

Cross-sectional SEM (Fig. 9(a)) shows complete filling with no observable voids over the full via depth. EDS mapping results (Fig. 9(c)-9(f)) show Cu and C distributed throughout the via cross-section, with no evidence of phase separation or axial particle settling. The uniform diamond distribution supports the effectiveness of the two-cycle injection approach and indicates that the electroless anchoring step maintains particle position during electroplating.

According to Fig. 10, the Si signal drops sharply at both via boundaries, confirming the integrity of the glass-via interface with no measurable elemental interdiffusion into the surrounding substrate. Within the via interior, the Cu and C signals exhibit alternating local maxima. This is consistent with

a heterogeneous Cu-diamond composite microstructure. Discrete diamond particles are individually embedded within a continuous copper matrix rather than segregated into distinct layers.

IV. RESULT AND DISCUSSION

The Cu–diamond composite thermal vias demonstrated here are fabricated using a route that differs from established bulk Cu–diamond composite methods. Prior approaches, such as high-temperature high-pressure sintering, spark plasma sintering, vacuum hot pressing, and melt infiltration, all rely on either applied consolidation pressure or an unconstrained part geometry. Those requirements do not translate to interposer-level heat management. In particular, the 100 µm-diameter, 1000 µm-deep TGV is a rigid, high-aspect-ratio channel with

glass sidewalls that are both thermally and mechanically mismatched to the metal. This process boundary condition is incompatible with pressure-driven sintering and with high-temperature liquid-phase infiltration inside the via. Under these constraints, electroless-assisted electroplating provides a practical in-via consolidation mechanism.

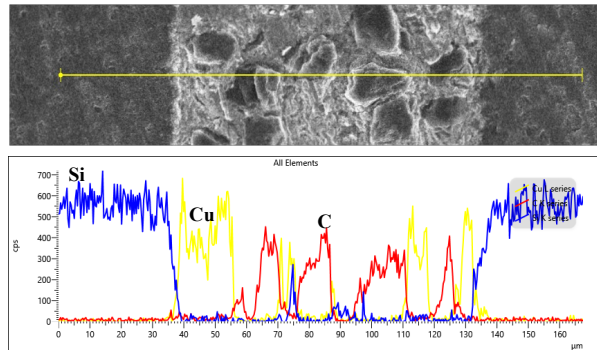


Fig. 10 Line-scan microstructural and compositional characterization of the Cu-diamond composite within the TGV.

Achieving void-free filling at a 10:1 aspect ratio required coordinated control of both plating and particle loading. Single-sided electroplating in deep vias is known to produce closure voids near the mid-depth region. Here, dual-sided electroplating was used to promote symmetric growth from both via entrances. This strategy drives the final closure plane toward the via center when the current distribution is balanced between the two faces. In addition, the two-cycle diamond injection strategy increased particle packing and improved through-depth uniformity. By replenishing particles between cycles, the process reduces large electrolyte-filled gaps that can persist during early-stage plating when the particle network is sparse. The EDS results show a uniform co-distribution of Cu and C without detectable segregation, which defines a process window for subsequent thermal performance optimization.

V. CONCLUSION

This paper reports the first Cu-diamond composite thermal vias fabricated in a glass core material. High-aspect-ratio TGVs (100 μm diameter, 1000 μm depth; 10:1 aspect ratio) were fully filled with a Cu-diamond composite using 20 μm TiC-pre-coated diamond particles at a target loading of 50 vol%. The particle size was selected based on the Hasselman-Johnson model. The model gives a thermally effective lower bound of 9.1–13.2 μm and a geometric upper bound of 20 μm for the 100 μm via diameter, and it predicts an effective thermal conductivity of 438–514 $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ at the design point.

To enable composite consolidation inside confined vias, we developed a process that combines electroless Cu encapsulation, vacuum-assisted two-cycle particle injection, electroless anchoring, and dual-sided electroplating. This process achieved void-free filling with a uniform composition, which is not achievable using conventional bulk Cu-diamond fabrication routes. Cross-sectional SEM and EDS results confirm complete filling and a homogeneous co-distribution of Cu and diamond over the full via depth. These results demonstrate the feasibility of electroplating-based Cu-diamond composite integration in packaging-relevant via structures and provide a foundation for

thermal management in high-density 2.5D heterogeneous integration using glass interposers.

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REFERENCES

- [1] J. H. Lau, "Recent Advances and Trends in Chiplet Design and Heterogeneous Integration Packaging," *Journal of Electronic Packaging*, vol. 146, no. 1, p. 010801, Mar. 2024, doi: 10.1115/1.4062529.
- [2] Z. Chen, J. Zhang, S. Wang, and C.-P. Wong, "Challenges and prospects for advanced packaging," *Fundamental Research*, vol. 4, no. 6, pp. 1455–1458, Nov. 2024, doi: 10.1016/j.fmre.2023.04.014.
- [3] J. Cho et al., "Modeling and Analysis of Through-Silicon Via (TSV) Noise Coupling and Suppression Using a Guard Ring," *IEEE Trans. Compon., Packag. Manuf. Technol.*, vol. 1, no. 2, pp. 220–233, Feb. 2011, doi: 10.1109/TCPMT.2010.2101892.
- [4] V. Sukumaran et al., "Design, Fabrication, and Characterization of Ultrathin 3-D Glass Interposers With Through-Package-Vias at Same Pitch as TSVs in Silicon," *IEEE Trans. Compon., Packag. Manuf. Technol.*, vol. 4, no. 5, pp. 786–795, May 2014, doi: 10.1109/TCPMT.2014.2303427.
- [5] A. B. Shorey and R. Lu, "Progress and application of through glass via (TGV) technology," in 2016 Pan Pacific Microelectronics Symposium (Pan Pacific), Big Island, HI, USA: IEEE, Jan. 2016, pp. 1–6, doi: 10.1109/PanPacific.2016.7428424.
- [6] Y.-H. Chen et al., "20" x 20" Panel Size Glass IPD Interposer Manufacturing," in 2016 IEEE 66th Electronic Components and Technology Conference (ECTC), Las Vegas, NV, USA: IEEE, May 2016, pp. 2146–2150, doi: 10.1109/ECTC.2016.400.
- [7] Y. Kim, "Electrical Performance Analysis of High-Speed Interconnection and Power Delivery Network (PDN) in Low-Loss Glass Substrate-Based Interposers," *Micromachines*, vol. 14, no. 10, p. 1880, Sep. 2023, doi: 10.3390/mi14101880.
- [8] Y. Yang, K. E. Lahaie, and T. Wei, "Double-Sided Copper Filling of Small Diameter, High-Aspect Ratio Through-Glass Vias in High-Density Glass Interposers," *IEEE Trans. Compon., Packag. Manuf. Technol.*, vol. 15, no. 7, pp. 1529–1537, Jul. 2025, doi: 10.1109/TCPMT.2025.3557232.
- [9] S. Cho, V. Sundaram, R. R. Tummala, and Y. K. Joshi, "Impact of Copper Through-Package Vias on Thermal Performance of Glass Interposers," *IEEE Trans. Compon., Packag. Manuf. Technol.*, vol. 5, no. 8, pp. 1075–1084, Aug. 2015, doi: 10.1109/TCPMT.2015.2450731.
- [10] H.-C. Chien, C.-H. Chien, M.-J. Dai, R.-M. Tain, W.-C. Lo, and Y.-J. R. Lu, "Thermal Characteristic and Performance of the Glass Interposer with TGVs (Through-Glass Via)," *International Symposium on Microelectronics*, vol. 2013, no. 1, pp. 000611–000617, Jan. 2013, doi: 10.4071/isom-2013-WP11.
- [11] E. Fey, J. Li, and N. Dimitrov, "Fast and Cost-Effective Superconformal Filling of High Aspect Ratio through Glass Vias Using MTT Additive," *J. Electrochem. Soc.*, vol. 164, no. 6, pp. D289–D296, 2017, doi: 10.1149/2.0741706jes.
- [12] P. Ogutu, E. Fey, and N. Dimitrov, "Superconformal Filling of High Aspect Ratio through Glass Vias (TGV) for Interposer Applications Using TNBT and NTBC Additives," *J. Electrochem. Soc.*, vol. 162, no. 9, pp. D457–D464, 2015, doi: 10.1149/2.0641509jes.
- [13] C. H. Jung, J. P. Jung, A. Sharma, and H.-S. Kim, "Advanced through-glass via (TGV) electro-filling and solder bumping for miniaturized 3D MEMS packaging," *Journal of Alloys and Compounds*, vol. 1038, p. 182619, Aug. 2025, doi: 10.1016/j.jallcom.2025.182619.
- [14] Y.-H. Chang, Y.-M. Lin, C.-Y. Lee, P.-C. Hsu, C.-M. Chen, and C.-E. Ho, "Through glass via (TGV) copper metallization and its microstructure modification," *Journal of Materials Research and Technology*, vol. 31, pp. 1008–1016, Jul. 2024, doi: 10.1016/j.jmrt.2024.06.137.

- [15] S. Lyu, T. Beechem, and T. Wei, "Reliability and Microstructure Characterization of Through-Silicon Vias (TSV) at Different Aspect Ratios Using EBSD-Raman Spectroscopy," in 2025 IEEE 75th Electronic Components and Technology Conference (ECTC), Dallas, TX, USA: IEEE, May 2025, pp. 380–385. doi: 10.1109/ECTC51687.2025.00071.
- [16] M. Wang et al., "Effect of heat treatment processes on the Cu-electrodeposited through glass vias (TGV) plate," *Journal of Alloys and Compounds*, vol. 997, p. 174928, Aug. 2024, doi: 10.1016/j.jallcom.2024.174928.
- [17] S. Q. Jia and F. Yang, "High thermal conductive copper/diamond composites: state of the art," *J Mater Sci*, vol. 56, no. 3, pp. 2241–2274, Jan. 2021, doi: 10.1007/s10853-020-05443-3.
- [18] Y. Xue et al., "Research progress in interface optimization and preparation technology of high thermal conductivity diamond/copper composite materials," *Front. Mater.*, vol. 12, p. 1582990, Apr. 2025, doi: 10.3389/fmats.2025.1582990.
- [19] K. Yoshida and H. Morigami, "Thermal properties of diamond/copper composite material," *Microelectronics Reliability*, vol. 44, no. 2, pp. 303–308, Feb. 2004, doi: 10.1016/S0026-2714(03)00215-4.
- [20] J. Peng, F. Zhang, Y. Zhou, L. Xiong, Y. Huang, and H. Tang, "Fabrication of diamond/copper composite thin plate based on a single-layer close packed diamond particles network for heat dissipation," *Chemical Engineering Journal*, vol. 476, p. 146666, Nov. 2023, doi: 10.1016/j.cej.2023.146666.
- [21] J. Shen et al., "Review of recent research on thermal management of diamond/metal composites," *Materials Science and Technology*, vol. 41, no. 7, pp. 488–511, May 2025, doi: 10.1177/02670836241261451.
- [22] S. Arai and M. Ueda, "Fabrication of high thermal conductivity copper/diamond composites by electrodeposition under potentiostatic conditions," *J Appl Electrochem*, vol. 50, no. 5, pp. 631–638, May 2020, doi: 10.1007/s10800-020-01414-3.
- [23] H. J. Cho, Y.-J. Kim, and U. Erb, "Thermal conductivity of copper-diamond composite materials produced by electrodeposition and the effect of TiC coatings on diamond particles," *Composites Part B: Engineering*, vol. 155, pp. 197–203, Dec. 2018, doi: 10.1016/j.compositesb.2018.08.014.
- [24] Y. Wu et al., "Microstructure of Cu-diamond composites with near-perfect interfaces prepared via electroplating and its thermal properties," *Materials Characterization*, vol. 150, pp. 199–206, Apr. 2019, doi: 10.1016/j.matchar.2019.02.018.
- [25] Y. Luo et al., "High thermal conductive Cu-diamond composite sheets reinforced with single-layer diamond prepared by a three-step electroplating process," *Diamond and Related Materials*, vol. 152, p. 111984, Feb. 2025, doi: 10.1016/j.diamond.2025.111984.
- [26] Y. Wang and B. Sun, "Anomalous strong size effect on thermal conductivity of diamond microparticles," *Applied Physics Letters*, vol. 125, no. 4, p. 042202, Jul. 2024, doi: 10.1063/5.0211873.
- [27] L. G. Min, H. Kwon, M. Asheghi, and K. E. Goodson, "Visualization of Spatial Thermal Conductivity and Boundary Conductance in Diamond Microparticles," *IEEE Trans. Compon., Packag. Manufact. Technol.*, pp. 1–1, 2026, doi: 10.1109/TCPMT.2025.3646970.
- [28] Z. Abdallah, J. W. Pomeroy, E. Neubauer, and M. Kuball, "Thermal Characterization of Metal-Diamond Composite Heat Spreaders Using Low-Frequency-Domain Thermoreflectance," *ACS Appl. Electron. Mater.*, vol. 5, no. 9, pp. 5017–5024, Sep. 2023, doi: 10.1021/acsaelm.3c00771.
- [29] Y. Yang, J. Chien, S. Lyu, and T. Wei, "Development of Straight, Small-Diameter, High-Aspect Ratio Copper-Filled Through-Glass Vias (TGV) for High-Density 3D Interconnections," in 2025 IEEE 75th Electronic Components and Technology Conference (ECTC), Dallas, TX, USA: IEEE, May 2025, pp. 1036–1042. doi: 10.1109/ECTC51687.2025.00181.